

REVIEW

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Horse gram: an under-utilized nutrient-rich legume crop as a potential candidate for sustainable agriculture

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Abstract Nutrient deficient foods are widely recognized as a major cause of malnutrition and also impact global food security. This has increased focus on nutrient-rich, underutilized food crops as a potential solution. *Macrotyloma uniflorum* (Horse gram) is an important yet neglected legume, rich in proteins, carbohydrates, vitamins, fatty acids, fiber, minerals, and valuable phytochemicals, making it useful for both human and animal consumption. Horse gram is mentioned in various traditional food systems, underscoring its cultural significance. The crop can withstand harsh environmental conditions such as drought, and low soil fertility. Considering the nutritional benefits of horse gram, there is a need to develop conservation strategies and value-added products to boost its market potential as a sustainable solution against global food insecurity and malnutrition. This review provides an insight into horse gram's botanical features, nutritional profile, and development of value-added products. The overall aim is to highlight the benefits of horse gram and encourage its value addition, and utilization to address malnutrition, enhance food security, and support sustainable agriculture.

Clinical trial number Not applicable.

Keywords Food security, Malnutrition, Zero hunger, Sustainable agriculture, Product development

Background

Out of the approximately 3,74,000 known plant species that have been accepted worldwide, more than 50,000 plant species contribute to edible plants including 7,039 wild species [1, 2]. India alone comprises 8% of the world's plant species, with 1,403 wild edible plants belonging to 184 families out of the 45,000 globally

recorded species [3, 4]. Despite such abundance, there are only 15 food crops that serve 90% of the human population's energy intake and only 3 staple crops, including maize, wheat and rice, serving more than 4 billion people across the world [5].

The 2023 Global hunger Index (GHI) data report a moderate score of 18.3, with undernourishment rising to 735 million people. Malnutrition and micronutrient deficiencies result from diets high in calories but lacking essential nutrients. Utilizing underexploited edible crops could support a sustainable food system and help achieve SDG – 1 (Zero hunger) by 2030 [6].

Macrotyloma uniflorum, commonly known as horse gram, is an under-utilized, nutrient-dense legume of the Fabaceae family [7]. It is a herbaceous, annual plant rich in proteins, carbohydrates, fatty acids, vitamins, minerals, and phytochemicals such as phenols, flavonoids and

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anthocyanins [8, 9]. Adapted to drought and poor soil conditions, horse grams' resilience makes it suitable for rainfed agriculture in arid or drought-prone regions [11]. Its nutritional profile and environmental tolerance position it as a potential candidate for addressing malnutrition and enhancing food security through sustainable agriculture in developing countries [11, 12]. Traditionally, seeds are soaked before cooking, as authentic cuisines are known to improve digestion and provide a nutritious diet. The leaves are also consumed in some regions as a rich source of vitamins and minerals [10, 13].

Despite its nutritional properties, the crop has been overlooked in favor of more commercially viable crops due to low market demand and limited consumer awareness. Factors such as shorter shelf-life caused by pest infestation, and the presence of nutrient-limiting compounds that impair mineral bioavailability, further hinder its cultivation and utilization. Therefore, developing advanced processing techniques and product innovation is critical to promote its cultivation and consumption.

This review emphasizes the significance of increasing awareness about the nutritional profile and health-promoting properties of horse gram, advocating for enhanced research aimed at agronomic yield optimization and nutritional enhancements. Such efforts are intended to facilitate greater adoption and commercial exploitation through value-added processing.

Main text

Horse gram: habit and habitat

The name *Macrotyloma* originates from a combination of Greek words, “makros”, “tylos” and “loma”, meaning “larger knob”, and “margin”, respectively referring to *Macrotyloma* as “knobby stature on pods” [14]. Horse gram is an annual perennial crop (Fig. 1) that thrives in soils like clay loam paddy, deep red loam and black cotton soil with an annual temperature requirement of 18–27 °C [7]. It tolerates 40 °C, but grows more slowly below 20 °C [15]. Reaching an average height of 30–40 cm, it has a weak and slender stem that makes it grow like a bush [16]. Tilling, sowing, and nutrient management methods significantly improve horse gram production [17]. The crop is cultivated worldwide in subtropical, temperate, and semi-arid areas at an elevation of 1800 m above sea level (asl) or higher, and on low or marginal fertility soil with rainfall as low as 380 mm [18]. It is mostly cultivated for food and fodder in Africa, South-Eastern Asia and Australia [19]. Crop residue and straw are used as green manure, fodder and fuel booster.

Nutritional profile

Horse gram is a rich source of carbohydrates, protein, fats, fibers, vitamins and minerals and nutrients like iron, molybdenum and calcium [20]. Its nutritional profile is influenced by factors including genotype, soil, fertilizer application, weather and climatic conditions, cultural practices, postharvest handling and storage [11]. Despite

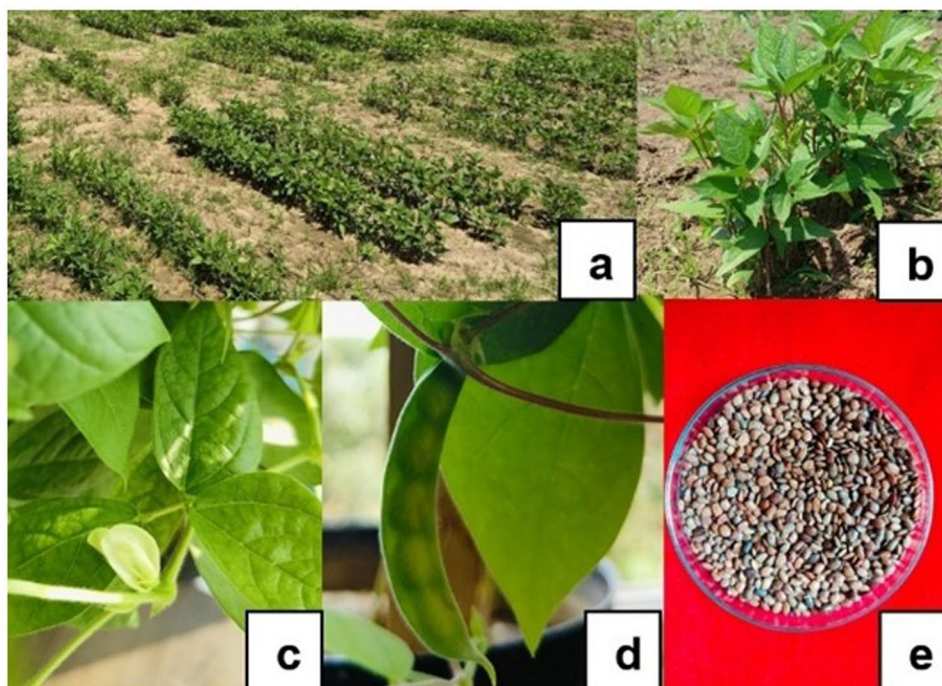


Fig. 1 Morphological features of horse gram (a. Horse gram germplasm established in the field; b. Horse gram plant; c. Horse gram flower; d. Horse gram pod; e. Horse gram seeds)

its high nutritional value, its consumption remains majorly limited to rural farming communities and low-income populations, rendering it an underutilized legume [21]. Several studies have attempted to quantify the nutrients in horse gram seeds. The carbohydrates in whole and dehulled horse gram seeds are known to vary from 51.9 to 60.9% and 56.8–66.4%, respectively and include soluble sugars like glucose, inositol, arabinose, galactose, fructose, sucrose and maltose [22, 23]. Raw seeds contain 22–44% starch (about 85% digestible starch, 14.47% resistant starch with 3.38% linked with insoluble dietary fibers) [24]. The protein content in horse gram seeds varies from 18.5 to 31.2%, depending on the cultivar [25]. Dehulled seeds exhibit higher protein levels (18.4–25.5%) compared to whole seeds (17.9–25.3%). The wild variety *Macrotyloma sargarhwalensis*, has a higher protein content ($38.4 \pm 1.03\%$) as compared to other commonly grown cultivars [26]. Globulin constitutes 60–90% of total protein content, with glutelin and albumin each accounting for 10–20% [27]. Horse gram makes a good complement to a cereal-based diet due to higher lysine content (0.52 g g^{-1} of N) than black gram (0.48 g g^{-1} of N) and pigeon pea (0.40 g g^{-1} of N) and surpasses kidney beans in several amino acids: threonine (3.8 mg/100 g), leucine (9 mg/100 g), iso-leucine (6.1 mg/100 g), phenylalanine (6.3 mg/100 g), valine (6.5 mg/100 g) and histidine (3.2 mg/100 g), compared to lower levels in kidney beans [7, 28]. These attributes position horse gram as a valuable plant-based protein source. The fat content in horse gram ranges from 0.6% to 2.6%, with dehulled seeds containing 0.8 to 2.1%, and whole seeds exhibiting 0.7 to 2.1% crude fat [20]. The seeds are rich in essential fatty acids, comprising 27.5% saturated and 72.5% unsaturated fatty acids, including linoleic, oleic, and linolenic acids [20, 29]. Total dietary fiber content is 28.8%, predominantly insoluble fiber (27.8%) and soluble fiber (1.1%), with an IDF: SDF ratio of 24.6. Conversely, horse gram flour contains 16.3% dietary fiber—14.9% insoluble, 1.4% soluble, and 2.2% resistant starch [30]. The fiber exists as beta-glucan, pectin, hemicellulose, and cellulose [31]. Moisture content in seeds ranges from 6.5% to 10.8%, and ash content varies from 3.4% to 4.3%, indicating mineral richness [25, 32, 33]. Horse gram is also an exceptional source of minerals, notably calcium, iron, zinc, magnesium, phosphorus and potassium [34]. The leaves contain about 4.5% mineral content, higher than that of other vegetables, which range from 1.5 to 2.4% [35]. The seeds also provide vitamins such as niacin, pantothenic acid, riboflavin, pyridoxine, thiamine, tocopherol, and folic acid, which are essential for physiological functions; deficiencies of which may cause diseases such as night blindness, rickets, and beriberi [36]. Nutritional profiles of different plant parts and

comparative analyses with other legumes are summarized in Tables 1 and 2.

Phytochemical profile

Horse gram is also a rich source of phytochemicals. The seeds contain about 0.9 mg/g of total phenols, predominantly p- coumaric acid ($40.9 \mu\text{g/g}$ dry weight dw) and p-hydroxy benzoic acid with $20.1 \mu\text{g/g}$ dw). The cotyledons are rich in phenolics, including, flavonoids, such as ferulic acid, p-hydroxybenzoic acid (7.8 mg/100 dw), quercetin ($129.5 \text{ mg/100 g dw}$), myricetin (35.5 mg/100 g dw) and kaempferol ($117.2 \text{ mg/100 g dw}$). The embryonic axes have been reported to contain isoflavonoids—daidzein (22.2 mg/100 g) and genistein (44.7 mg/100 g) [42, 43, 44]. Tannin concentrations range from 763.7 to 895.9 mg/ 100 g, exceeding those in other legumes [34]. Phytic acid, or inositol hexaphosphate, constitutes the primary phosphorus reservoir, with levels of 184 mg/100 g in seeds (accounting for approximately 57% of total phosphorus) and about 10.2 mg/g in flour [16]. The structures of these phenolic compounds, flavonoids, and tannins are depicted in Fig. 2.

Nutrient-limiting compounds and processing methods to mitigate their effect

Pulses contain compounds called nutrient-limiting factors, such as alkaloids, saponins, tannins, phenols, phytic acid, and protein inhibitors. Horse gram seeds specifically contain amylase and trypsin inhibitors, and phytates. The flour prepared from seeds has been found to contain phytic acid ($10.2 \pm 0.4 \text{ mg/g}$), polyphenols ($14.3 \pm 0.4 \text{ mg GA/g}$), oligosaccharides (26.8 mg/g), and trypsin inhibitor (9246 ± 18 trypsin units inhibited TIU/g) [45]. These compounds, while offering some health benefits—including reduced risk of colon cancer, constipation, gallstones, heart disease, diverticulosis, dental caries and diabetes, significantly impair mineral bioavailability [46]. Phytates, tannins and oxalic acid chelate minerals like iron and calcium, forming insoluble salts that hinder absorption, despite the high mineral content of horse gram [47]. Other antinutritional factors include proteinase and trypsin inhibitors, flatulence factors and hemagglutinins. Protease inhibitors hinder protein digestion by targeting proteolytic enzymes, thereby impacting the nutritional quality [23]. Trypsin inhibitors, predominantly abundant in horse gram compared to other legumes, significantly block the trypsin activity [48]. Oligosaccharides such as verbascose, stachyose, and raffinose— α -galactosyl sucrose derivatives in horse gram can cause gastrointestinal discomfort [49]. Hemagglutinins (lectins and phytoagglutinins), mainly located in seeds, but also present in roots, stems and leaves [50], may induce intestinal damage, nutritional anemia by interacting with immune factors and erythrocytes [51]. Proper processing reduces

Table 1 Complete nutritional profile of horse gram

S. No.	Major nutritional component	Part studied	Different forms of major nutritional component	Concentration	Major nutritional component	References
1.	Carbohydrates	Whole seeds Dehulled seeds	Monosaccharide Disaccharide oligosaccharide	51.9–60.9% 56.8–66.4%	Starch (22–45% of total carbohydrate content)	[37]
3.	Dietary fiber	Seeds Flour	Soluble form, Insoluble form Soluble form, Insoluble form	Total dietary fiber: 28.80% Soluble dietary fiber: 1.13% Insoluble dietary fiber: 27.82% Total dietary fiber: 16.3% Soluble dietary fiber: 1.4% Insoluble dietary fiber: 14.9%	Cellulose Hemicellulose pectin beta-globulin	[30]
3.	Protein	Seeds	-	Total: 18-28.5% which includes the following Cotyledon part: 89% Seed coat: 10% Embryonic part: 1%	Globulin followed by glutelin and albumin	[38, 39]
4.	Amino acid	Seeds	Methionine Tryptophan Cysteine Histidine Threonine Iso leucine Phenyl alanine Valine Argentine Leucine	1.16 mg/ 100 gm 1.16 mg/ 100 gm 1.96 mg/ 100 gm 3.16 mg/ 100 gm 3.82 mg/ 100 gm 6.14 mg/ 100 gm 6.31 mg/ 100 gm 6.47 mg/ 100 gm 8.8 mg/ 100 gm 8.96 mg/ 100 gm	-	[7]
5.	Minerals	Before germination After germination	Zinc Iron Magnesium Calcium Phosphorus Potassium Iron Magnesium Calcium Phosphorus Potassium	3.4 mg/ 100 gm 6.77 mg/ 100 gm 172 mg/ 100 gm 287 mg/ 100 gm 311 mg/ 100 gm 762 mg/ 100 gm 4.03 mg/ 100 gm 136 mg/ 100 gm 11.6 mg/ 100 gm 256 mg/ 100 gm 675 mg/ 100 gm	Calcium Iron	[40]
6.	Fat	Seeds (Whole seeds contain 0.70–2.06% while dehulled seeds contains 0.81–2.11%)	Unsaturated fatty acids Linolenic acid Oleic Linoleic Saturated fatty acids Myristic Stearic Arachidic Palmitic	72.49% 13.56% 16.15% 42.78% 27.5% 0.36% 2.32% 2.85% 21.97%	-	[20]
7.	Vitamins	-	Niacin Pantothenic acid Riboflavin Pyridoxine Thiamine Tocopherol Folic acid	1.82 mg/ 100 gm 1.58 mg/ 100 gm 0.24 mg/ 100 gm 0.21 mg/ 100 gm 0.38 mg/ 100 gm 0.30 mg/ 100 gm 165 mg/ 100 gm	-	[36]

Table 2 Comparative analysis of nutritional profiles of some legume crops with horse gram [41]

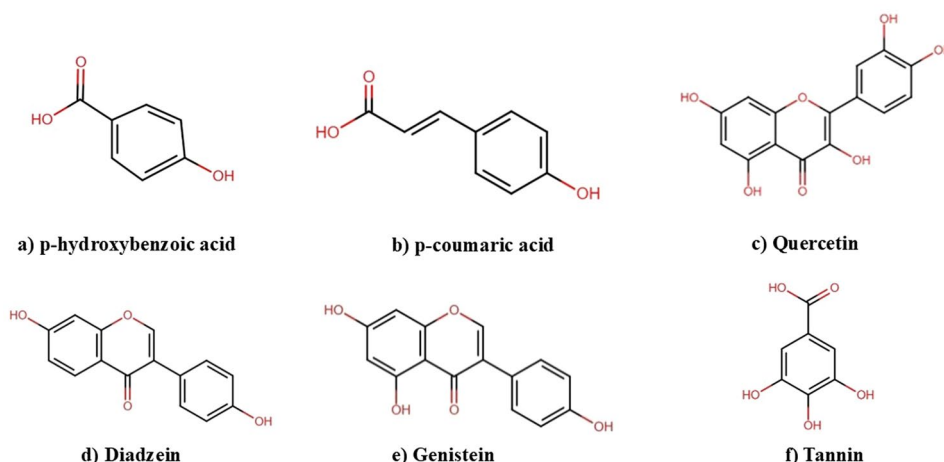
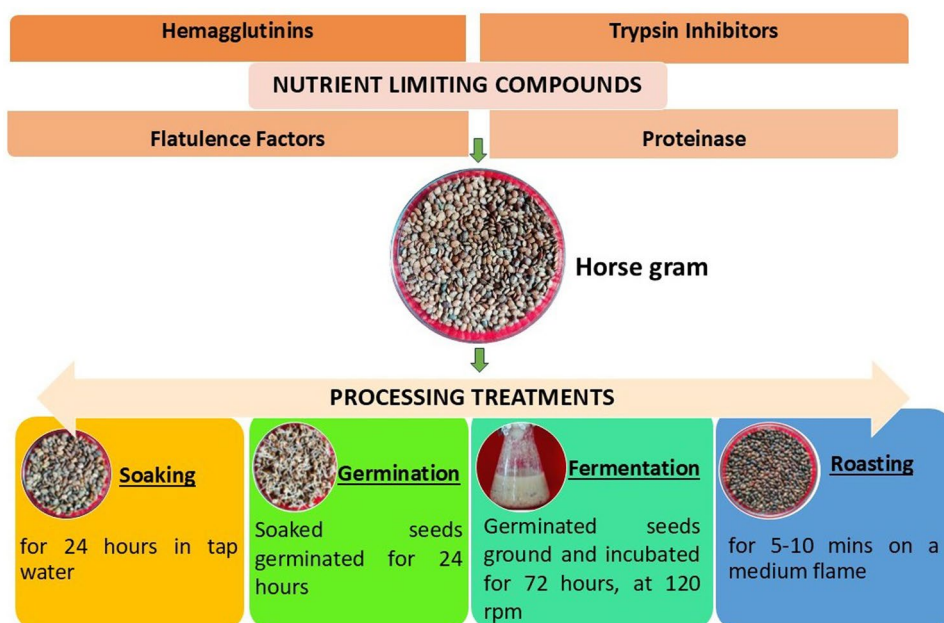
Pulses	Energy (KCal)	Protein (g)	Fat (g)	Carbohydrate (g)	Total dietary fiber (%)
Chickpea	368	21	5.7	61	22.7
Pigeon pea	342	21.7	1.49	62	15.5
Urd bean	347	24	1.6	63.4	16.2
Mung bean	345	25	1.1	62.6	16.3
Lentil	346	27.2	1	60	11.5
Field pea	345	25.1	0.8	61.8	13.4
Kidney bean	345	23	1.3	63.4	18.2
Cowpea	346	28	1.3	63.4	18.2
Moth bean	330	24	1.5	61.9	-
Horse gram	321	23.6	2.3	59.1	15

these antinutritional factors, improving nutritional bioavailability.

Processing methods

The nutrient-limiting constituents in horse gram restrict its nutritional efficacy by affecting bioavailability, absorption and digestibility, thereby decreasing its dietary utilization [52, 53]. Processing techniques such as soaking, roasting, cooking, dehiscing, and germination effectively reduce these antinutrients, enhancing both acceptability and nutritional profile, as illustrated in Fig. 3 [32].

Soaking A domestic hydration process lasting 12–18 h, reduces trypsin inhibitors, tannins, and phytic acid, while augmenting mineral content (like manganese, zinc, cop-

**Fig. 2** Structures of phenolics (p-hydroxybenzoic acid; p-coumaric acid); Flavanols (Quercetin; Daidzein; Genistein) and Tannin**Fig. 3** Types of processing treatments to reduce the effects of nutrient-limiting factors in horse gram

per, and iron). Soaking for 12 and 24 h has been shown to lower trypsin inhibitors from 12.2 to 11.7 mg/100 g, phytic acid from 10.4 to 9.2 mg/100 g, and tannins from 2.3 to 2.1 mg/100 g [32].

Germination A cost-effective, rapid method that reduces tannins, flavonoids, phenols, post-sprouting [26, 54]. It enhances moisture, ash and total protein content, while slightly decreasing carbohydrates, crude fat and fiber [26]. Enzymatic activation during germination hydrolyzes starches, fats, and proteins, decreasing antinutritional compounds, including trypsin inhibitors [32].

Dehulling Removal of seed hulls reduces oxalic acid, trypsin inhibitors, phytic acid and tannins, concurrently increasing protein, lipid, and sugar content. The process may decrease the levels of antioxidant compounds, such as flavonoids, thereby affecting their antioxidant properties. When combined with germination, dehulling synergistically enhances horse gram's nutritive value and reduces antinutrients [53].

Fermentation Treatment with sodium hypochlorite followed by 48-hour fermentation decreases pH from 6.4 to 4.6, with reductions in phytate, oxalate, and tannins by 69.5%, 66.8%, and 69.4%, respectively. Phytate degradation is mediated by phytase activity from lactic acid bacteria at low pH; oxalate and tannins are utilized and hydrolyzed during fermentation [52].

Roasting Roasting dried seeds at 160 °C in an iron pan for 10 min reduces tannins, flavonoids, polyphenols, phytates, and oxalis by 28%, 48.7%, 28.3%, 22.4%, and 22.45%, respectively. These reductions result from thermal denaturation and insoluble complex formation; flavonoids and polyphenols decline due to thermal degradation [52, 55].

Impact of value-addition to horse gram-based food products: opportunities for achieving sustainable development goals

Value addition is a vital process for enhancing the marketability of agricultural commodities by improving product quality and brand recognition, thereby facilitating increased income generation and employment opportunities [56]. It enables year-round accessibility and prolongs shelf-life, effectively reducing post harvest losses. Additionally, value addition can strengthen off-farm employment, promote technology transfer, and augment workforce capacity, ultimately contributing to economic development of a nation [57, 58]. The extent of value addition can be quantified by comparing the costs associated with raw material acquisition and inputs to the returns from the processed, value added products [59].

Horse gram presents a viable candidate for value-addition through biofortification to enhance its nutritional and phytochemical profiles, enabling the development of diverse horse gram-based food products. Several companies have developed various horse gram-based food products. Table 3 provides a list of some globally reported food items made from horse gram, that can be seamlessly integrated into daily dietary practices.

Research gaps and methodological limitations

Research on horse gram has primarily focused on its nutritional composition, phytochemical properties, and applications in value-added food products, establishing the crop as a rich source of protein, dietary fibers, and minerals [25, 26, 34]. Studies have also investigated the effects of processing treatments, including dehulling, soaking, roasting, germination, and fermentation, on improving nutrient bioavailability and reducing antinutritional factors like phytates and tannins [23, 32]. Furthermore, the incorporation of horse gram into diverse food products has demonstrated considerable potential for nutritional enhancement, particularly through increased protein and fiber content [62, 64]. Despite the growing body of research, several important knowledge gaps continue to hinder the effective utilization and commercialization of horse gram. Existing studies are frequently constrained by limited germplasm representation, small sample sizes, and a lack of standardized methodologies for nutritional and phytochemical analysis, making comparisons across studies difficult. Moreover, the influence of environmental and agronomic factors on nutrient composition and bioactive compounds remains inadequately understood. Research on shelf-life stability, storage characteristics, industrial-scale processing feasibility, consumer acceptance based on sensory attributes is also limited. Furthermore, many of the proposed health-promoting and functional properties of horse gram require stronger mechanistic and clinical validation. By critically synthesizing the available evidence, this review identifies key research priorities and provides a comprehensive framework for future investigations. In doing so, it enhances the current understanding of horse gram and supports its broader utilization in sustainable agriculture, functional food development, and nutrition-sensitive food systems.

Future directions

To address the inconsistencies in the recent literature on horse gram, there is a need to shift towards standardized, integrative research frameworks. Future studies should focus on developing uniform analytical protocols to evaluate the horse grams's nutritional and phytochemical richness, and rely more on advanced technologies to address climate change. These include studies such

Table 3 Horse gram-based food products reported along with their corresponding nutritional composition

S. No.	Category	Product name	Nutritional values	Quantity	References
1.	Fermented food	Fortified Bun(6% germinated horse gram flour bread)	Fiber	1.23%	[60]
			Fat	3.74%	
			Protein	9.8%	
			Calcium	128 mg/100g	
			Iron	4.07 mg/100g	
			Oxalate	0.7 mg/g	
			Tannin	2.06 mg/g	
			Phytate	2.46 mg/g	
			Ash	1.36%	
			Moisture	36.05%	
2.	Bakery products	Kaulath (fermented product which was prepared using <i>Penicillium camemberti</i>)	Carbohydrate	73.4%	[61]
			Protein	19.6%	
			Ash	4.1%	
			Fat	1.8%	
		Cookies(20gm of horse gram was added to 80 gm of wheat flour)	Iron	4.67 gm	[62]
			Calcium	2.44 gm	
			Carbohydrate	62 gm	
			Protein	12.50 gm	
			Fat	22.32 gm	
			Insoluble fiber	1.28 gm	
			Soluble fiber	7.25 gm	
		Cookies(0.3% guar gum; 20% pearl millet flour; 30% horse gram flour; 50% refined wheat flour)	Protein	15.28%	[63]
			Fat	22.74%	
			Carbohydrate	52.65%	
			Moisture	4.32%	
			Crude fiber	2.51%	
			Ash	2.50%	
		Biscuits	Carbohydrates	62.85%	[64]
			Fat	16.17%	
			Protein	10.23%	
			Moisture	7.06%	
			Fiber	1.83%	
			Calcium	134 mg/100g	
			Iron	2.63 mg/100g	
			Zinc	1.05 mg/100g	
			Manganese	0.61 mg/100g	
			Copper	0.48 mg/100g	

Table 3 (continued)

S. No.	Category	Product name	Nutritional values	Quantity	References
3.	Ready-to-eat	Idly-mix powder	Carbohydrate	170.2 gm	[65]
			Protein	18 gm	
			Fiber	6.5 gm	
			Iron	8 mg	
			Ascorbic acid	1.5 mg	
			Phosphorus	355 mg	
			Calcium	290 mg	
			Moisture	7%	
			Ash	3.4%	
		Nachos	Carbohydrates	52.91%	[66]
			Fats	22.40%	
			Protein	17.80%	
			Fiber	6.82%	
			Calcium	0.221%	
			Iron	0.08%	
		Instant Dosa mix	Carbohydrates	39.12%	[13]
			Protein	15.33%	
			Fiber	3.12%	
			Moisture	4.12%	
			Calcium	146 mg/100g	
			Iron	1.8 mg/100g	
4.	Sweets	Kasar	Phytic acid	0.80 mg/100g	[67]
			Carbohydrates	36.34%	
			Protein	6.95%	
			Fiber	1.92%	
			Ash	1.46%	
			Fat	20.88%	
5.	Beverages	Horse gram milk	Iron	3.72 mg/100g	[68]
			Total sugar	1.35-2.72 g/100g	
			Reducing sugar	0.79-2.30 g/100g	
			Starch	0.67-1.79 g/100g	
			Protein	16.05-38.68g/100g	
			Tannin	22.65-46.80 mg/100g	
			Oxalate	38.39-82 mg/100g	
			Phenols	8.63-20GAE mg/100g	
			Ascorbic acid	2.79-6.61 mg/100g	

Table 3 (continued)

S. No.	Category	Product name	Nutritional values	Quantity	References
6.	Others	Adai(Ragi flour along with 60% roasted horse gram flour)	Iron	5.57 gm	[62]
			Calcium	1.47 gm	
			Carbohydrate	60.12 gm	
			Protein	10.79 gm	
			Fat	19.42 gm	
			Fiber	4.38 gm	
			Insoluble fiber	0.30 gm	
			Soluble fiber	4.07 gm	
		Rasam mix(made using germinated horse gram seeds)	Carbohydrates	46.7%	[13]
			Protein	28.34%	
			Moisture	5.89%	
			Fiber	5.89%	
			Calcium	256 mg/100g	
			Iron	6.44 mg/100g	
			Tannins	3.3 mg/100g	
			Phytic acid	3 mg/100g	
		Chutney powder	Carbohydrates	44.56%	[13]
			Protein	26.78%	
			Fiber	4.98%	
			Moisture	3.22%	
			Calcium	291.5 mg/100g	
			Iron	6 mg/100g	
			Moisture	8.14%	[69]
			Carbohydrates	74.67%	
			Protein	10.5%	
			Fiber	2.21%	
			Fat	1.07%	
			Calcium	97 mg/100g	
			Iron	5.09 mg/100g	
			Copper	0.44 mg/100g	
			Manganese	0.8 mg/100g	
			Zinc	1.38 mg/100g	
		Mathri	Carbohydrates	75.53%	[67]
			Protein	13.48%	
			Fiber	3.50%	
			Ash	2.07%	
			Fat	21.87%	
			Iron	6.24 mg/100g	
			Carbohydrates	74.06%	[69]
			Protein	12.25%	
			Moisture	8.2%	
			Fiber	1.91%	
			Fat	1.69%	
			Manganese	0.92 mg/100g	
			Calcium	60 mg/100g	
			Iron	4.4 mg/100g	
			Zinc	1.21 mg/100g	
			Copper	0.46 mg/100g	

as GWAS (genome-wide association studies) to identify potential cultivars with reduced nutrient-limiting factors and increased protein digestibility. This will also help reveal the mechanisms behind drought, salt, and heavy

metal stress tolerance and should promote the breeding of climate-resilient cultivars capable of withstanding rising temperatures and reduced precipitation due to climate change. Additionally, product development should

support expansion into novel biofortified plant-based formulations and integration with other pulse-based products while improving commercial scalability, product stability and consumer acceptance. Consequently, this crop's inherent resilience, rich nutritional and phytochemical profile, advanced processing approaches, and clinical validation will position it as a key component in addressing food security and climate resilience.

Conclusion

Horse gram is an ancient, nutrient-dense legume crop rich in bioactive compounds with potential applications in the food and pharmaceutical sectors, thus contributing to public health. Efforts need to focus on enhancing yield, quality, and genotypic and phenotypic traits, as well as on germplasm conservation. The crop's underutilization is primarily due to nutrient-limiting factors; however, processing techniques and breeding low-nutrient-inhibiting cultivars can mitigate these issues. Further research is also needed to identify its bioactive constituents, and elucidate the mechanisms underlying their effects on human health. A sustainable production of horse gram would require integrated approaches in conservation, standardized nutritional profiling, product development, consumer acceptance and market infrastructure to realize its full potential.

Abbreviations

GHI	Global hunger index
IDF	Insoluble dietary fiber
SDF	Soluble dietary fiber
GAE	Gallic acid equivalent

Author contributions

Aditi Sharma: Methodology, Investigation and Writing Original Draft; Manu Pant: Conceptualization, Project Administration, Review and Editing; Anjali Rawat: Investigation and Writing Original Draft; Rajnish Rawat: Resources and Investigation; Priyanshi Bisht: Resources and Methodology; Sanjay Kumar: Supervision, Writing-Review and Editing.

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Data availability

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Ethics declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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